

Learning Disciplinary Domain Distributions for Swedish University Courses: A Multi-Label Study with KB-BERT

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Abstract

University courses in Sweden are assigned disciplinary domains that determine state funding allocations. In this project, we examine automatic classification of Stockholm University courses as both a multi-label and label distribution learning task, using course plans and metadata from a 2023 Ladok extract. We compare TF-IDF-based linear models with a fine-tuned KB-BERT model, evaluating both domain set prediction and percentage-based domain distributions. KB-BERT significantly outperforms TF-IDF baselines in multi-label classification, achieving a subset accuracy of 90.8% compared to 82.9% for the strongest baseline (Linear SVC). Paired bootstrap resampling confirms the statistical significance of this improvement. For distribution prediction, direct training with a KL-divergence loss yields a 48% lower mean absolute error than post-hoc normalization of binary predictions (1.44 vs. 2.79 percentage points). Topic-model-based error analysis, carried out together with a domain expert, indicates that most large discrepancies between predictions and gold labels reflect administrative conventions, program constraints, or historical policies rather than content misunderstanding. These findings suggest that content-based models can support administrators by highlighting courses whose classifications merit review.

1 Introduction

University courses are assigned to disciplinary domains for purposes such as internal planning, national statistics, and state funding allocation. In Sweden, these domains—called *utbildningsområden* (UO), determine how much funding universities receive per student and credit. Misclassification, therefore, has direct financial consequences and can distort how resources are distributed among faculties.

At Stockholm University (SU), course classifications are decided by administrators and subject experts following national regulations and local guidelines. This is a complex task since many courses are interdisciplinary by nature and classified by percentage-based distributions (e.g., 70% humanities, 30% natural sciences). From a machine-learning perspective, this constitutes a multi-label and label-distribution text classification problem: given the textual description of a course, the system should predict one or more disciplinary labels and their relative proportions.

Common multi-label classification (MLC) methods include problem-transformation methods such as one-vs-rest training, as well as algorithm-adaptation methods that extend classical classifiers to handle multiple labels (Herrera et al., 2016). MLC typically treats each label as binary: either present or absent.

This limitation is addressed by Label Distribution Learning (LDL), a framework where each label has an associated *degree* describing how strongly it applies to an instance (Geng, 2016). The SU course-classification task, with its explicit percentage allocations, aligns with this method.

Recent work has shown that transformer-based models achieve strong performance on text classification tasks, including multi-label problems (Tarekegn et al., 2024). In this project, we finetune a Swedish language transformer-based model KB-BERT, pretrained on a large corpus by the National Library of Sweden (Malmsten et al., 2020). This model has been successfully adapted for text classification tasks, such as sentiment analysis (Hägglöf, 2023) and topic modelling (Fano and Haf-fenden, 2022).

We compare traditional TF-IDF-based linear models with a fine-tuned KB-BERT model on SU course descriptions, and evaluate both multi-label performance and the ability to predict full percentage-based domain distributions. For distri-

bution prediction, we consider two variants: (i) a baseline that first produces independent sigmoid scores for each label and then (post-hoc) converts them into a distribution by normalising the scores to sum to one, and (ii) a label-distribution model with a softmax output layer trained directly to match the gold distributions using a KL-divergence loss.

Concretely, we address the following research questions:

1. How accurately can machine learning methods reproduce existing course classifications, and how do TF-IDF-based approaches compare to transformer-based models (KB-BERT)?
2. To what extent can models predict the percentage-based distribution of disciplinary domains, and does direct distributional learning outperform post-hoc normalization of binary predictions?
3. What systematic patterns characterize prediction errors, and what do these patterns reveal about the relationship between course content and administrative classification?

Based on prior work demonstrating the effectiveness of pretrained language models for text classification (Tarekegn et al., 2024), and the identification of our task as a case of LDL (Geng, 2016), we formulate the following hypotheses:

- H1** Fine-tuned KB-BERT will significantly outperform TF-IDF-based linear classifiers on multi-label classification metrics.
- H2** Direct distributional prediction will yield more accurate percentage allocations than post-hoc normalization of binary predictions.

2 Data

2.1 Corpus

The corpus was provided by SU Administration as a 2023 extract from Ladok, the national student information system used by Swedish universities. It contains course plans with metadata including titles, descriptions, learning objectives, and disciplinary domain classifications. Decision dates for included course versions span from 2005 to early 2024.

Each course is classified into one or more of $K = 10$ disciplinary domains (“utbildningsområden”, UO), which determine resource allocation under the Swedish higher education funding model. Multi-label courses have percentage allocations summing to 100%.

2.2 Label Distribution

The label distribution is somewhat imbalanced. The three largest education areas—Humanities (HU, 2434), Social Sciences (SA, 2442), and Natural Sciences (NA, 2441)—appear in approximately 35%, 34%, and 19% of course versions, respectively. Other domains are less frequent: Other (ÖV, 2447) accounts for about 8%, while Law (JU, 2436), Teacher Education (LU, 2438), Medicine (ME, 2439), Technology (TE, 2444), and Health Care (VÅ, 2445) each occur in roughly 4–5%. Widening Participation (VU, 2451) is the rarest label, accounting for fewer than 2%. Most courses are assigned to a single education area: 82% have just one UO label. In total, there are 25 distinct label combinations, and multi-label cases are relatively rare and dominated by Social-Sciences combinations (e.g. SA+TE, SA+ME, SA+VÅ, SA+LU).

2.3 Preprocessing

The raw corpus contains 12,949 rows in long format (one row per UO label assignment). A *course version* is defined as a unique (course ID, decision date) pair, since both content and classifications may change over time. Aggregating UO codes and percentages by course version leaves us with 9,770 instances, representing 4,880 unique courses.

Text fields (title, description, learning objectives) were concatenated and normalized: removing markup, collapsing whitespace, and stripping special characters.

2.4 Train-Validation Split

To prevent information leakage, all versions of the same course are assigned to the same split. We perform a stratified split at the course level (by primary UO), then propagate assignments to all corresponding versions. This yields 7,865 training instances (3,904 courses) and 1,905 validation instances (976 courses).

3 Methodology

This section describes the experimental setup used to address the research questions. We frame the

task in two complementary ways: as multi-label classification (predicting which UO codes apply) and as distribution prediction (estimating percentage allocations). Two model families are compared: TF-IDF-based classifiers and fine-tuned KB-BERT.

3.1 Problem Formulation

Given a course description text x , the goal is to predict an assignment to one or more of $K = 10$ disciplinary domains. The corpus provides two forms of supervision: (1) a binary vector $\mathbf{y} \in \{0, 1\}^K$ indicating which domains are assigned, and (2) a probability vector $\mathbf{p} \in [0, 1]^K$ where $\sum_k p_k = 1$, derived from percentage allocations.

3.2 Baseline Models

We employ TF-IDF features with character n-grams (length 3–5, word boundaries) combined with one-vs-rest linear classifiers: **Logistic Regression** with balanced class weights, and **Linear SVC**. Both produce binary predictions for each UO code.

3.3 Transformer Models

We fine-tune KB-BERT (Malmsten et al., 2020), a Swedish BERT model from the National Library of Sweden. Two training objectives are explored:

Binary Multi-Label Classification. A classification head with sigmoid activation is trained using binary cross-entropy loss. Each domain is treated as an independent binary classification task. For distribution prediction, the sigmoid outputs (not constrained to sum to 1) are post-hoc normalized to sum to 1.

Direct Distribution Prediction. A softmax output layer trained with KL divergence loss directly models the percentage allocation as a probability distribution:

$$\mathcal{L} = D_{\text{KL}}(\mathbf{p} \parallel \hat{\mathbf{p}}) = \sum_k p_k \log \frac{p_k}{\hat{p}_k} \quad (1)$$

This Label Distribution Learning formulation directly optimizes for percentage allocations.

3.4 Training Details

Transformer models are fine-tuned for 3 epochs using AdamW (learning rate 2×10^{-5} , weight decay 0.01), batch size 8, and a maximum sequence length of 512 tokens.

3.5 Evaluation

Multi-Label Metrics. We report:

- **subset accuracy** (exact match of all labels)
- **micro-F1** (global F1 across all assignments)
- **macro-F1** (unweighted average per-label F1)
- **Hamming loss** (fraction of incorrect labels).

Distribution Metrics. We report:

- **mean absolute error** (MAE) in percentage points
- **top-1 accuracy** (primary domain match),
- **cosine similarity**
- **Jensen-Shannon distance** (JSD).

3.6 Statistical Testing

To determine whether differences between systems are statistically significant, we employ paired nonparametric bootstrap resampling over course versions (Efron and Tibshirani, 1994). For each comparison, we repeatedly sample N course versions with replacement, recompute the evaluation metric for both systems on each resampled set, and record the difference in scores. We use $B = 10,000$ bootstrap samples and form 95% percentile confidence intervals from the empirical distribution of differences. A two-sided p-value is estimated as the proportion of bootstrap differences whose sign is opposite to the observed difference (doubled for a two-sided test); differences are considered significant at $\alpha = 0.05$.

3.7 Error Analysis

To investigate systematic patterns (RQ3), we employ a two-stage analysis validated through expert consultation with SU Administration.

Data Exploration. LDA topic modeling (Blei et al., 2001) ($K = 10$) is used to discover courses where topics diverge from UO labels: For each course, its topic distribution is compared to the typical pattern for its assigned UO label(s).

Model Error Analysis. High-MAE predictions from the BERT distributional model (>10 percentage points) were analyzed to identify systematic prediction errors. Cases were interpreted with the help of the domain expert to distinguish genuine model errors from administratively-driven disagreements.

Topic entropy $H = -\sum_k \theta_k \log \theta_k$ was compared between single-label and multi-label courses to assess whether multi-label assignments reflect content diversity.

Table 1: Multi-label classification results. All BERT-SVC differences significant at $p < 0.001$.

Metric	LogReg	SVC	BERT
Subset Acc.	77.8	82.9	90.8
Micro-F1	87.8	90.0	93.6
Macro-F1	83.6	85.8	90.5
Hamming ↓	3.02	2.31	1.54

Table 2: Distribution prediction results. All differences significant at $p < 0.001$.

Metric	Binary	Distrib.
Top-1 Acc.	85.3	92.4
MAE (pp) ↓	2.79	1.44
Cosine	0.938	0.951

4 Results

4.1 Multi-label Classification (RQ1)

Table 1 presents classification performance. KB-BERT achieves subset accuracy of 90.76% compared to 82.94% for Linear SVC and 77.80% for Logistic Regression.

The improvement of KB-BERT over Linear SVC is significant across all metrics ($p < 0.001$). Subset accuracy improves by 7.82 percentage points (95% CI [6.30, 9.34]), micro-F1 by 3.57 points, and macro-F1 by 4.69 points. Hamming loss decreases from 2.31% to 1.54%, a 33% relative reduction. These results confirm H1.

4.2 Distribution Prediction (RQ2)

Table 2 compares BERT training objectives for percentage prediction. The distributional model substantially outperforms binary classification with post-hoc normalization.

Top-1 accuracy improves from 85.25% to 92.39% ($\Delta = +7.14$ pp, $p < 0.001$). MAE drops from 2.79 to 1.44 percentage points, a 48% relative improvement. These results confirm H2: direct distribution prediction yields significantly more accurate allocations than post-hoc normalization.

4.3 Error Analysis (RQ3)

Data Exploration. LDA topic modeling identified courses where unsupervised topic assignments diverged from administrative labels. Expert consultation revealed that this divergence largely reflects administrative conventions in the gold standard. Three patterns emerged: (1) regula-

tory stipulations assigning ÖV to certain language courses for funding purposes (Regeringskansliet, 2024); (2) program restrictions preventing LU classification for didactics courses outside the teacher program; (3) historical policy fixing linguistics courses at 50/50 HU/NA due to a departmental merger. Several flagged courses have since been administratively reclassified, suggesting topic modeling can identify review candidates.

Model Errors Analysis of high-MAE BERT predictions confirmed that most large disagreements correspond to administrative patterns rather than content misunderstanding. The model predicts content-appropriate labels (e.g., HU for language courses) that conflict with administratively-determined gold labels. Genuine failures were rare: “Runforskning” and “Namnforskning” received technology predictions due to generic descriptions; “Mentalt innehåll” (Philosophy of Mind) was predicted as ME/SA instead of HU.

Topic Entropy No significant difference emerged between single-label ($H = 1.56$) and multi-label ($H = 1.54$) courses, suggesting multi-label assignments reflect administrative partitioning rather than content diversity.

5 Discussion

5.1 Interpretation of Results

Both hypotheses were confirmed with strong statistical significance. The superiority of KB-BERT over TF-IDF baselines was expected and confirms recent trends in NLP showing that pretrained transformers capture semantic nuances that bag-of-words representations miss (Tarekegn et al., 2024). The 7.8 percentage point improvement in subset accuracy suggests that contextual embeddings are particularly valuable when distinguishing between related disciplinary domains (e.g., SA vs. HU for interdisciplinary social science courses).

The advantage of direct distributional learning over post-hoc normalization (H2) supports the theoretical motivation of Label Distribution Learning

5.2 Limitations

(Geng, 2016). Binary classification treats labels independently and cannot learn trade-offs between competing categories; the softmax formulation forces the model to allocate probability mass, better matching the structure of percentage-based as-

signments. The 48% reduction in MAE demonstrates that this architectural choice has a substantial practical impact.

5.3 Content Understanding vs. Shortcut Learning

This finding is reinforced by the error analysis: the model consistently predicts content-appropriate labels even when they conflict with administratively-determined gold labels. For example, language courses assigned to ÖV for regulatory funding reasons are predicted as HU based on their humanities content. This suggests the model has learned actual content representations.

5.4 Implications for Practice

Error analysis confirmed that most large prediction errors reflect administrative conventions, namely regulatory stipulations, program restrictions, and historical policy, rather than content misunderstanding. This implies that the model may be more useful for *content-based classification review* than for replicating existing administrative assignments. Courses where model predictions diverge strongly from gold labels are candidates for human review, as some may reflect outdated classifications. Indeed, several courses flagged by our LDA analysis have since been administratively reclassified.

5.5 Limitations

Several limitations apply. First, the corpus includes only courses from Stockholm University, so generalization to other contexts is untested. Second, data from 2023 was used due to technical issues with accessing current Ladok records. Third, potential overlap exists between course descriptions and KB-BERT’s pretraining corpus (which includes official government documents), risking inflated performance (Malmsten et al., 2020). Fourth, a single train-validation split was used, limiting variance assessment. Fifth, only 18% of courses were multi-label, so distributional predictions focus on edge cases. The error analysis was qualitative, based on 50 high-divergence cases. While expert review suggested many “errors” reflect administrative conventions, the relative influence of content versus policy on classifications was not quantified. High accuracy indicates effective learning of the classification scheme, but the balance between content- and rule-based patterns remains unclear. Future work

could evaluate generalization to other Swedish universities, compare advanced MLC architectures (e.g., Tarekegn et al. (2024)) to our softmax baseline, or conduct longitudinal studies to analyze administrative decision-making dynamics.

6 Ethical Considerations

The corpus contains administrative metadata and public course descriptions but no personal data about students or instructors; course information is available via the university catalogue available through university course catalogues (SU, 2025). Automated classification of courses into funding-relevant domains could, in principle, be misused to optimize allocations rather than accurately reflect content. We therefore view our models as decision-support tools to assist human reviewers, not as replacements for expert judgment. The finding that many “errors” reflect administrative rules rather than content misunderstanding also underscores the importance of transparency: gold-standard labels encode both subject matter and policy decisions, and stakeholders should be aware of this when interpreting model outputs.

7 Conclusion

We addressed the task of automatically classifying Swedish university courses into disciplinary domains, framing it as both multi-label classification and label distribution learning. Fine-tuned KB-BERT significantly outperformed TF-IDF baselines on multi-label metrics ($p < 0.001$), and direct distributional training yielded 48% lower MAE than post-hoc normalization of binary predictions.

Error analysis, conducted through LDA topic modeling and expert consultation, revealed that most prediction-gold disagreements reflect administrative conventions—regulatory funding rules, program restrictions, and historical policy—rather than failures of content understanding. This suggests that content-based models may be valuable for identifying courses whose classifications merit review.

The Label Distribution Learning framework provides a natural fit for tasks where labels have associated proportions. Future work should evaluate generalization to other institutions and explore integration of structured metadata alongside text features.

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Code and Data Availability

All code used for this study will be made available at <https://github.com/fboglind/lis070-su-admin-project>. Information about data access will also be provided in the repository.

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A Evaluation Metrics

A.1 Multi-Label Classification Metrics

Let $\mathbf{y} \in \{0, 1\}^K$ be the gold label vector and $\hat{\mathbf{y}} \in \{0, 1\}^K$ the predicted label vector for K labels across N instances.

Subset Accuracy. The proportion of instances where all predicted labels exactly match all gold labels:

$$\text{Subset Acc} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\mathbf{y}_i = \hat{\mathbf{y}}_i] \quad (2)$$

Micro-F1. F1-score computed globally by aggregating true positives, false positives, and false negatives across all labels:

$$\text{Micro-F1} = \frac{2 \cdot \text{TP}}{2 \text{ TP} + \text{FP} + \text{FN}} \quad (3)$$

Macro-F1. Unweighted average of per-label F1 scores:

$$\text{Macro-F1} = \frac{1}{K} \sum_{k=1}^K \text{F1}_k \quad (4)$$

Hamming Loss. The fraction of labels incorrectly predicted:

$$\text{Hamming} = \frac{1}{NK} \sum_{i=1}^N \sum_{k=1}^K \mathbf{1}[y_{ik} \neq \hat{y}_{ik}] \quad (5)$$

A.2 Distribution Prediction Metrics

Let $\mathbf{p} \in [0, 1]^K$ be the gold distribution and $\hat{\mathbf{p}} \in [0, 1]^K$ the predicted distribution, both summing to 1.

Mean Absolute Error (MAE). Average absolute difference between predicted and gold percentages, reported in percentage points:

$$\text{MAE} = \frac{100}{NK} \sum_{i=1}^N \sum_{k=1}^K |p_{ik} - \hat{p}_{ik}| \quad (6)$$

Top-1 Accuracy. Whether the predicted primary domain matches the gold primary domain:

$$\text{Top-1 Acc} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\arg \max_k p_{ik} = \arg \max_k \hat{p}_{ik}] \quad (7)$$

Cosine Similarity. Mean cosine similarity between predicted and gold distribution vectors:

$$\text{Cosine} = \frac{1}{N} \sum_{i=1}^N \frac{\mathbf{p}_i \cdot \hat{\mathbf{p}}_i}{\|\mathbf{p}_i\| \|\hat{\mathbf{p}}_i\|} \quad (8)$$

Jensen–Shannon Distance. The reported Jensen–Shannon distance, is computed as the square root of the Jensen–Shannon divergence using `scipy.spatial.distance.jensenshannon` with natural logarithm:¹

$$\text{JS dist}(\mathbf{p}, \hat{\mathbf{p}}) = \sqrt{\frac{1}{2} D_{\text{KL}}(\mathbf{p} \parallel \mathbf{m}) + \frac{1}{2} D_{\text{KL}}(\hat{\mathbf{p}} \parallel \mathbf{m})} \quad (9)$$

where $\mathbf{m} = \frac{1}{2}(\mathbf{p} + \hat{\mathbf{p}})$ is the mixture distribution.

¹Note that this differs from JSD proper; the distance is $\sqrt{\text{JSD}}$ and ranges from 0 to approximately 0.833 with natural log, rather than $[0, 1]$ with log base 2.